

CSE48D-OBJECT ORIENTED PROGRAMMING USING JAVA



NAME-SUJAL KUMAR

ROLL.NO-16

SECTION-JK231

PROBLEM STATEMENT

- Develop a traffic light simulation based project in java which can guide based upon lights which action need to be performed.

Objective

design and implement a Java-based project that simulates the behavior of a **traffic light system**, and based on the current light (RED, YELLOW, GREEN), the system should:

- **Guide the user or driver** on what action to take:
 - **RED** → STOP
 - **YELLOW** → Get Ready / Slow Down
 - **GREEN** → GO

JAVA PROGRAM

```
import java.util.Scanner;

class ManualTrafficLight {
    String currentColor = "RED";

    void changeLight() {
        if (currentColor.equals("RED")) {
            currentColor = "GREEN";
        } else if (currentColor.equals("GREEN")) {
            currentColor = "YELLOW";
        } else {
            currentColor = "RED";
        }
    }

    void showMessage() {
        if (currentColor.equals("RED")) {
            System.out.println("RED: Please STOP.");
        } else if (currentColor.equals("YELLOW")) {
            System.out.println("YELLOW: Get ready or SLOW down.");
        } else if (currentColor.equals("GREEN")) {
            System.out.println("GREEN: You can GO.");
        }
    }
}

class SmartTrafficLight {
    String currentColor = "RED";

    void setLightByTraffic(String traffic) {
        if (traffic.equalsIgnoreCase("high")) {
            currentColor = "RED";
        } else if (traffic.equalsIgnoreCase("medium")) {
            currentColor = "YELLOW";
        } else if (traffic.equalsIgnoreCase("low")) {
            currentColor = "GREEN";
        } else {
            System.out.println("Invalid traffic input.");
        }
    }

    void showMessage() {
        if (currentColor.equals("RED")) {
            System.out.println("RED: Please STOP.");
        } else if (currentColor.equals("YELLOW")) {
            System.out.println("YELLOW: Get ready or SLOW down.");
        }
    }
}
```

```

    } else if (currentColor.equals("GREEN")) {
        System.out.println("GREEN: You can GO.");
    }
}

}

public class TrafficLightS {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.println("Welcome to Traffic Light Simulation!");
        System.out.println("Choose mode:\n1. Manual\n2. Smart");
        int choice = scanner.nextInt();
        scanner.nextLine();

        if (choice == 1) {
            ManualTrafficLight light = new ManualTrafficLight();
            while (true) {
                light.showMessage();
                System.out.println("Press Enter to change the light or type 'exit' to
stop:");

                String input = scanner.nextLine();
                if (input.equalsIgnoreCase("exit")) {
                    break;
                }
                light.changeLight();
            }
        } else if (choice == 2) {
            SmartTrafficLight light = new SmartTrafficLight();
            while (true) {
                System.out.println("Enter traffic level (high, medium, low) or 'exit'
to stop:");

                String traffic = scanner.nextLine();
                if (traffic.equalsIgnoreCase("exit")) {
                    break;
                }
                light.setLightByTraffic(traffic);
                light.showMessage();
            }
        } else {
            System.out.println("Invalid choice. Please run the program again.");
        }

        System.out.println("Simulation ended. Goodbye!");
        scanner.close();
    }
}

```

Explanation

This Java program simulates a **Traffic Light** with two modes: **Manual** and **Smart**. The program uses **Object-Oriented Programming (OOP)** concepts like **classes**, **methods**, and **encapsulation**.

Classes in the Program

1. ManualTrafficLight Class

- **Purpose:** Simulates a traffic light where the light changes manually with user input.

Fields:

String currentColor = "RED";

- Starts with the RED light.

Methods:

- changeLight(): Changes the traffic light from RED → GREEN → YELLOW → RED, and repeats.
 - showMessage(): Displays a message based on the current light color:
 - RED: "Please STOP."
 - YELLOW: "Get ready or SLOW down."
 - GREEN: "You can GO."
-

2. SmartTrafficLight Class

- **Purpose:** Simulates a smart traffic light that adapts based on the traffic level entered by the user.

Fields:

String currentColor = "RED";

- Starts with RED.

Methods:

- setLightByTraffic(String traffic): Changes the light based on the traffic level input:
 - high → RED
 - medium → YELLOW
 - low → GREEN
- showMessage(): Displays the same messages based on the current light color.

Main Program Flow (TrafficLightS Class)

1. Scanner Setup:

Scanner scanner = new Scanner(System.in);

- Used to read user input.

2. User Choice:

System.out.println("Choose mode:\n1. Manual\n2. Smart");

```
int choice = scanner.nextInt();
```

- User chooses between **Manual** or **Smart** mode.

3. **Manual Mode:**

- Creates a ManualTrafficLight object.
- Loops continuously:
 - Shows the current light.
 - Waits for user input: press Enter to change the light, type 'exit' to stop.

4. **Smart Mode:**

- Creates a SmartTrafficLight object.
- Loops continuously:
 - Asks the user for the traffic level (high, medium, low).
 - Changes the light based on input.
 - Displays the message for the current light.

5. **Invalid Choice Handling:**

- If the user enters something other than 1 or 2, an error message is displayed.

6. **Program End:**

- scanner.close(); to close the scanner.
- Displays a goodbye message.

OUTPUT

```
Welcome to Traffic Light Simulation!
Choose mode:
1. Manual
2. Smart
1
RED: Please STOP.
Press Enter to change the light or type 'exit' to stop:

GREEN: You can GO.
Press Enter to change the light or type 'exit' to stop:

YELLOW: Get ready or SLOW down.
Press Enter to change the light or type 'exit' to stop:
exit
Simulation ended. Goodbye!
```

```
Welcome to Traffic Light Simulation!
Choose mode:
1. Manual
2. Smart
2
Enter traffic level (high, medium, low) or 'exit' to stop:
high
RED: Please STOP.
Enter traffic level (high, medium, low) or 'exit' to stop:
medium
YELLOW: Get ready or SLOW down.
Enter traffic level (high, medium, low) or 'exit' to stop:
low
GREEN: You can GO.
Enter traffic level (high, medium, low) or 'exit' to stop:
exit
Simulation ended. Goodbye!
```

OOPs Concepts Used in the Traffic Light Simulation (Java)

1. Class & Object

- Classes: ManualTrafficLight, SmartTrafficLight, TrafficLightS

- Objects are created using:
ManualTrafficLight light = new ManualTrafficLight();
SmartTrafficLight light = new SmartTrafficLight();

2. Encapsulation

- Data (currentColor) and related methods (changeLight(), setLightByTraffic(), showMessage()) are wrapped inside the class.
- Provides data hiding and protects the internal state.

3. Abstraction

- Complex logic is hidden inside methods.
- Users interact with simple method calls like light.changeLight() or light.setLightByTraffic("high") without knowing the internal logic.

4. Polymorphism (Conceptual Use)

- Currently not implemented.
- Can be added by creating a base class or interface (e.g., TrafficLight) and overriding methods like showMessage().

5. Inheritance (Can be Enhanced)

- Not used in this version.
- Could improve the design by using a common base class for both light types.

6. Control Structures

- if-else used to control traffic light behavior.
- while loop used to keep the simulation running until the user exits.